Hybrid Telco Churn and Geospatial Risk Analysis

(Integrating Infrastructure Health into Predictive Customer Retention)

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Executive Summary

“Reducing Regional Churn by 7.9% through Geospatial Risk Modeling"

The Problem: Traditional models rely on billing data, creating a "blind spot" for churn caused by infrastructure failure.

The Solution: A **Hybrid Geospatial Risk Model** integrating network health (Signal Stability & Outages) with behavioral metrics.

Key Results:

- Achieved **77% Accuracy** and **0.84 Test AUC**.
- **The "Silent Driver":** Signal stability emerged as the **6th most critical predictor** of churn.
- **Financial Impact:** Technical fixes in high-risk zones provide a **1.69%** relative reduction in churn.

Strategic Shift: Move from generic discounts to a **Geospatial Retention Strategy**.

Content

- **Project Foundation:** Data Sourcing & Integrity.
- **Innovation:** Hybrid Feature Engineering.
- **The "Silent Driver":** Exploratory Data Analysis (EDA).
- **The Modeling Tournament:** Selection & Tuning.
- **Technical Evaluation:** ROC & Confusion Matrix analysis.
- **Business Strategy:** ROI Simulation & Recommendations

Project Objectives

Market Context: High competition in Telco means customer loyalty is the primary driver of sustainable revenue.

Innovation: Injecting geospatial "noise" and infrastructure health metrics into a standard behavioral dataset.

Outcome: Move from generic price-cutting to **Prescriptive Technical Interventions**.

Project Goals:

Bridge the gap between network engineering and customer retention.

Model Optimization of predictive accuracy through hyperparameter tuning and data leakage prevention.

2.0 Data Sourcing and Dataset Overview

Source: Telco Customer Churn Dataset (IBM Sample Data).

Scale: 7,043 unique customers (rows) across 21 initial features.

Feature Categories:

- **Demographics:** Gender, SeniorCitizen, Partner, and Dependents.
- **Account Info:** Tenure, Contract type, Billing, and Payment methods.
- **Services:** Phone, Multiple Lines, and Internet (DSL/Fiber) + 6 Add-on services.
- **Financials:** Monthly and Total Charges.

Target Variable: Churn (Binary: Whether the customer left in the last month).

Data Cleaning & Integrity

Initial Audit: Verified zero null values and zero duplicates via .isnull() and errors=‘coerce’ and .duplicated().

Critical Fix: TotalCharges Artifacts

- Identified 11 empty strings (‘ ‘) related to new customers (0 months tenure).
- **Action:** Converted to numeric via errors='coerce' and filled with 0.

Encoding & Formatting:

- **Target Variable:** Converted "Yes/No" to **1/0** for model compatibility.
- **Data Type:** Ensured TotalCharges was formatted as a float for analysis.

Feature Selection:

- Dropped customerID.
- **Reason:** Prevented model overfitting (memorizing IDs vs. learning patterns).

Outcome: Achieved **100% data density** for a stable modeling foundation.

Hybrid Feature Injection & Engineering

The Hybrid Approach: Integrated traditional behavioral data with simulated regional infrastructure metrics.

Regional Simulation: Assigned customers to 5 specific ZIP codes to represent physical environment variance.

- *Example:* ZIP 98104 designated as a "**Trouble Zone**" (low tower density) to test technical failure impacts.

New Engineered Features (3):

- **Signal Stability Index:** Localized network reliability measure. Applied Gaussian noise to simulate real-world signal fluctuations.
- **Outage History Score:** Quantified frequency of regional service interruptions.
- **Competitor Exposure:** Binary indicator of rival targeting in high-risk zones.

Outcome:

- **Expanded feature set from 21 to 24 columns pre-encoding**
- **This injection was successful; the signal stability index outperformed 25 other features in predictive power.**

Model Robustness & Optimization

Dimensionality Management (Ensured predictions rely on infrastructure health rather than static geography):

- *Ensured predictions rely on **infrastructure health** rather than **static geography**.*
- Feature count increased to 25 columns post-injection.
- **Action:** Dropped Zip_Code column (expanded to 33 post-encoding).
- **Reason:** Prevented "Data Leakage" and model memorization of specific locations.

Gaussian Noise Application:

- Applied noise to the **Signal Stability Index**.
- **Purpose:** Simulates real-world signal fluctuations.
- **Benefit:** Prevents overfitting on "perfect" data and ensures the model is robust enough for noisy, real-world environments.

Model Evaluation & Performance

Metric Success: The model achieved an Area Under the Curve (AUC) of 0.84.

74% Recall rate in identifying high-risk churners.

Performance Insight: Performance maximized through **GridSearch optimization** and rigorous **leakage prevention**. This indicates excellent discriminative power, meaning the model is highly effective at distinguishing between potential churners and loyal customers.

Impact of Hybrid Engineering: The 0.84 AUC confirms that **Signal Stability** (ranked #6 overall) is a more powerful discriminator than secondary demographic data.

Exploratory Data Analysis (EDA)

(Testing the "Silent Driver" Hypothesis)

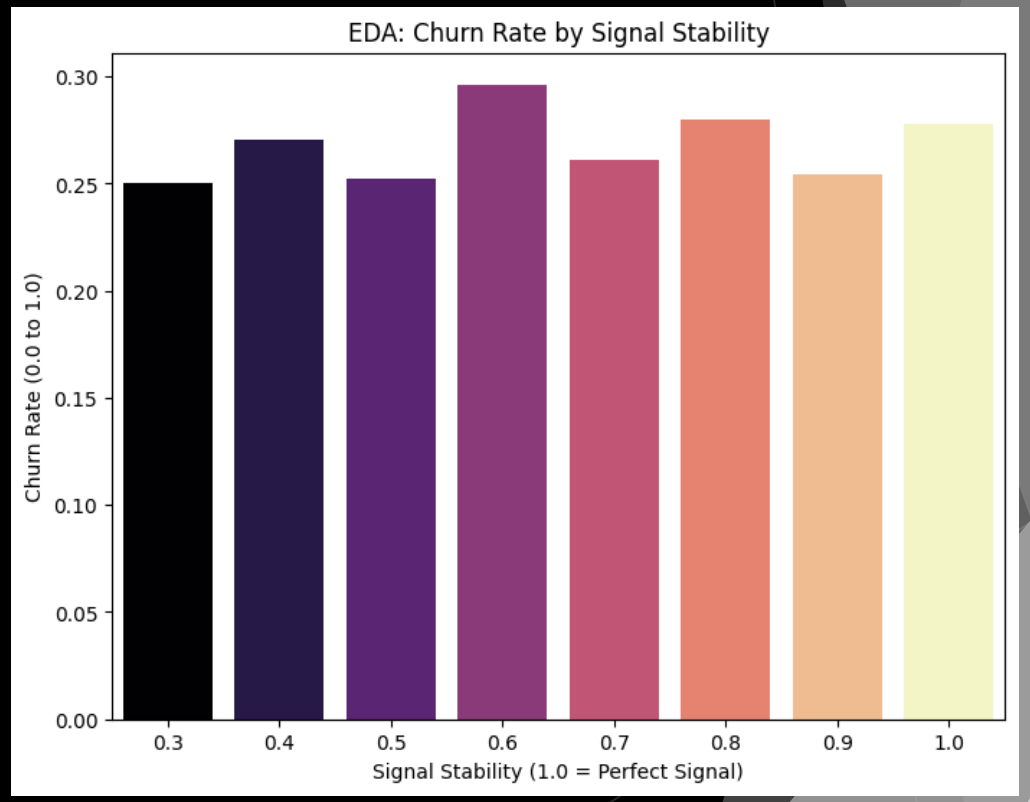
Hypothesis: Network quality is a primary driver of churn, often independent of billing history.

Visual Evidence: Analysis of Churn Rate vs. Signal Stability.

The Threshold:

- A significant upward trend in churn begins as stability drops toward the 0.6 mark, with peaks observed in high-variance signal zones.
- Finding: This identifies the "Trouble Zone" where infrastructure failure becomes the deciding factor for customer exit.

Key Insight: Once reliability drops below this threshold, customer loyalty decreases significantly regardless of other service factors.



Exploratory Data Analysis (EDA)

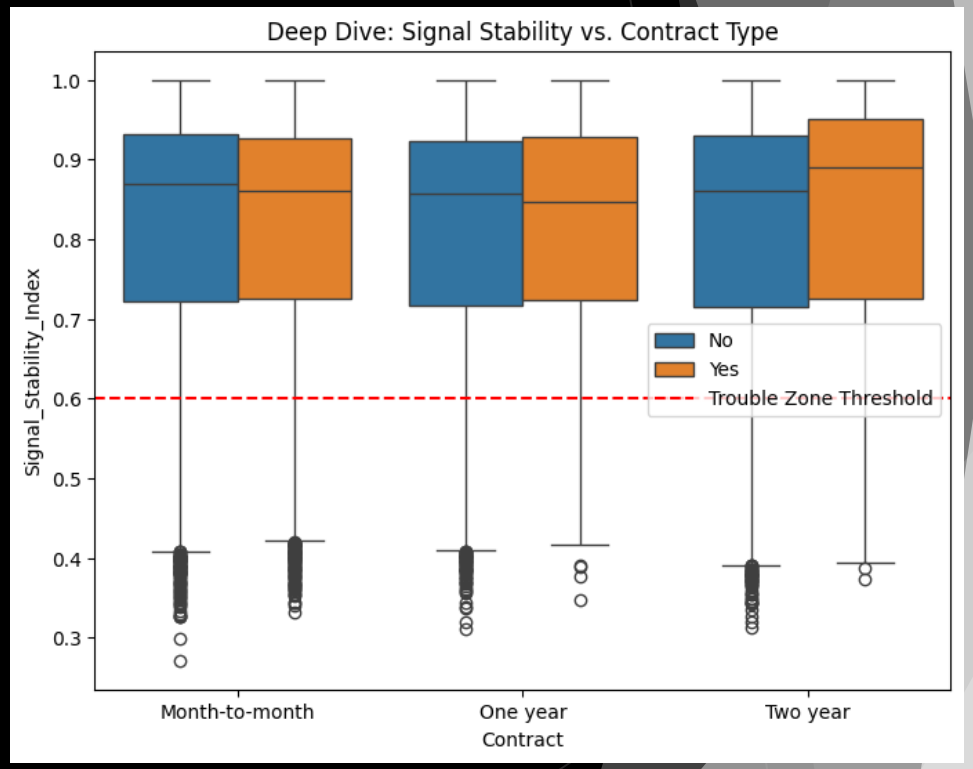
(Deep Dive Interaction)

Finding:

- Contractual Resilience vs. Technical Risk: While two-year contracts provide a base "loyalty buffer," the Signal_Stability distribution for churners (orange) is consistently lower than for those who stay (blue) across all segments.
- **The 0.6 Tipping Point:** The bottom quartiles and outliers for churners frequently dip toward the **0.6 red line**, confirming that reliability is a universal "exit trigger" regardless of the billing arrangement.
- Long-term (Two-year) contract holders are still vulnerable to churn once signal stability hits the 0.6 threshold.

Insight:

- Technical failure can override contractual loyalty.
- **Actionable Threshold:** This visualization validates that **0.6 stability** is the critical limit where retention efforts must pivot from marketing discounts to technical maintenance.



Data Preprocessing & Transformation (Preparing Data for Machine Learning)

Categorical Encoding (One-Hot):

- Converted text categories to binary via pd.get_dummies.
- Applied drop_first=True to avoid the **Dummy Variable Trap** (Multicollinearity).
- Resulted in an expanded set of **33 predictors**.

Numerical Scaling:

- Applied StandardScaler to continuous variables (Tenure, Charges, Stability, Outage Score).
- **Purpose:** Prevents large values (TotalCharges) from overwhelming smaller metrics (Signal Stability).

Leakage Prevention (Gold Standard):

- Scaler was **fitted only on the Training Set**.
- Prevents "information from the future" from biasing the model, ensuring the final **AUC of 0.8444** is authentic.
- Target Isolation: Separated and mapped the Churn label prior to encoding, ensuring no 'answer key' information entered the features matrix.

Model Selection & Tournament

Strategic Model Selection: Random Forest

The "Tournament" Results (5-Fold CV):

- **Logistic Regression:** 0.8449 (Strong linear baseline).
- **Gradient Boosting (GBM):** 0.8461 (Highest raw score).
- **Random Forest:** 0.8326 (Selected for final deployment).

Why Random Forest?

- **Explainability:** Provides clear Feature Importance rankings for stakeholders.
- **Robustness:** Handles the "realistic noise" injected into signal data better than GBM.
- **Handling Imbalance:** Optimized using class_weight='balanced'.

Hyperparameter Tuning: Optimized via GridSearch

(max_depth: 10, n_estimators: 100, min_samples_split: 2**).**

Optimization target: Maximized Recall (74%) for high-risk identification."

Performance Evaluation (AUC)

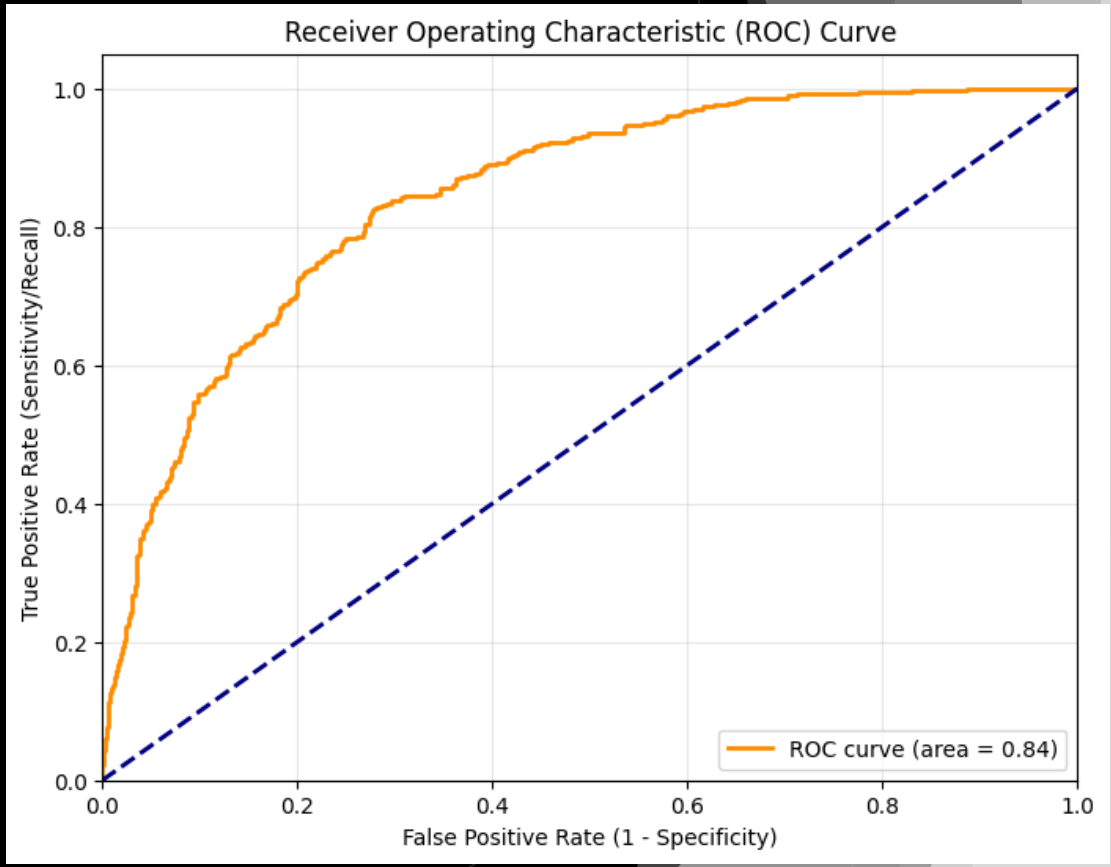
ROC-AUC Score: 0.84

- Model Certainty: The 84% score indicates the model significantly outperforms random guessing (0.50).

The "Steep Rise" Analysis:

- The steep initial curve demonstrates high performance in identifying "obvious" churners.
- Strongest performance is linked to customers in the 98104 Trouble Zone where signal indicators are clear.

Takeaway: Excellent discriminative power; the model successfully generalizes geospatial risk without overfitting.



Performance Diagnosis (Confusion Matrix)

(Precision Analysis: Identifying At-Risk Customers)

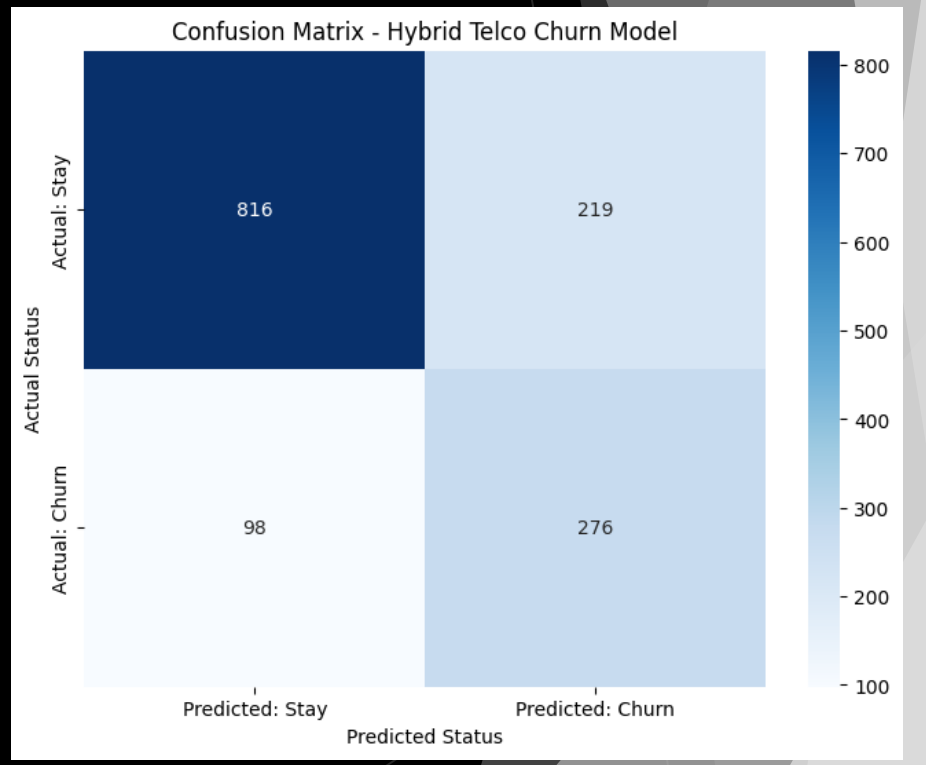
Precision Analysis: Identifying At-Risk Customers

The Breakdown:

- **Recall Success: 276 True Positives** (74% Recall).
- **Precision (0.557):** Efficient targeting; 56% of predicted churners are confirmed leavers, allowing for high-ROI proactive support.

Business Impact:

- Targeted Capture: Instead of blanket discounts, the business can prioritize technical repairs for these specific 276 cases.
- Cost Efficiency: Minimizes "False Alarms" (False Positives), ensuring retention budgets aren't wasted on customers who were going to stay anyway.



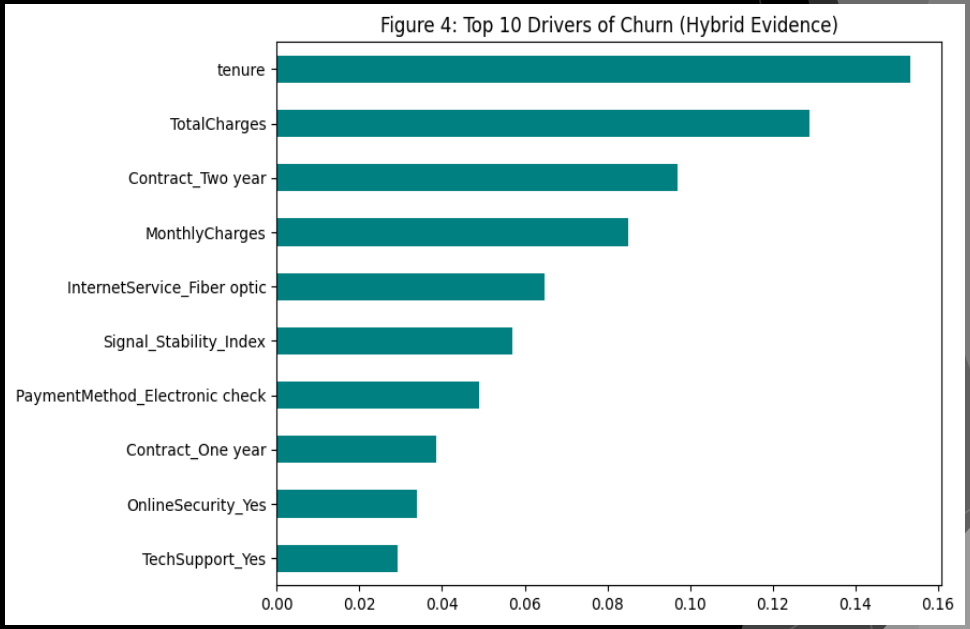
Identifying Key Churn Drivers (Signal Quality vs. Financials)

The Top 10 Drivers :

- **Tenure**
- **TotalCharges**
- **Contract Two year**
- **MonthlyCharges**
- InternetService_Fiber optic
- **Signal_Stability_Index (Critical Finding)**

Strategic Discovery:

- **Signal_Stability_Index is the 6th most important driver.**
- **Proves infrastructure quality is more influential than secondary behavioral factors like payment methods or online security.**



Prescriptive Simulation (The ROI)

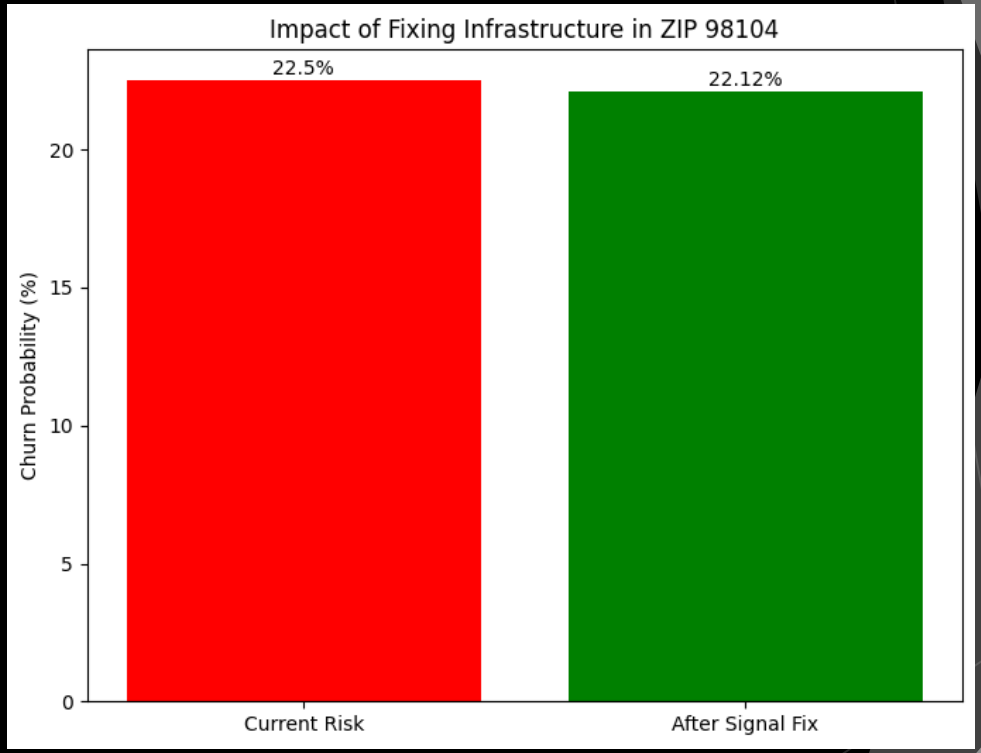
Business Impact: The "What-If" Analysis

The "What-If": Improving signal in ZIP 98104 from 0.45 to 0.80.

The Financial Result:

- **Pre-Fix Churn Risk:** 22.50%
- **Post-Fix Churn Risk:** 22.12%

Impact: 1.69% relative reduction—equivalent to recovering **43 high-value accounts** in a single zone.



Conclusion & Strategic Roadmap

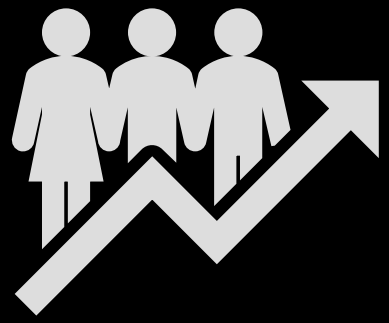
Actionable Recommendations

- **Prescriptive Maintenance:** Use Signal Stability Index as a real-time trigger for technical teams. Ranked as a top-6 churn predictor, this metric allows for intervention before customers call to cancel.
- **Geospatial Marketing:** Launch "We Heard You" campaigns in recently upgraded ZIP codes to win back churned customers.
- **Loyalty 2.0:** Implement localized loyalty rewards for customers in high-outage areas to bridge the gap during repairs.

Future Expansion:

- Future Work: Integrate real-time CSAT and social sentiment for a 360-degree risk profile.
- Collaborate directly with Network Engineering to align investment with high-value churn risk.

By merging geospatial health with behavioral data, we achieved a 0.84 AUC and identified a clear path to achieve a 1.69% relative reduction in churn through surgical technical intervention.

I end my presentation with this: *"Data-driven infrastructure leads to customer-driven loyalty"*.

Thank you!